

Week 2 Day 2 – LangChain Agent Write-up

Raw-Python Agent vs. LangChain Agent

In Day 1, I created an AI agent using raw Python. I had to manually manage the agent workflow and tool execution. In Day 2, I used LangChain to create an agent with tools, memory, and structured output. LangChain made the development process easier and more organized.

Comparison

Raw Python Agent	LangChain Agent
More steps had to be handled manually.	LangChain provides ready-made agent components.
Tool execution needed more manual code.	Tools can be registered and called by the agent.
Conversation history needed to be managed manually.	LangChain provides memory/history features.
Output formatting required extra work.	Pydantic can provide structured output.

What LangChain Made Easier

LangChain made it easier to connect the Gemini model with different tools. I created tools for addition, multiplication, and reading product prices from a CSV file. I also added conversation memory so the agent could use information from previous turns. Structured output made the final response more organized and predictable.

Annotated Tool Trace

User Question:

“What is the price of the Laptop, and then multiply that price by 2?”

Reason:

The agent identified that it first needed to find the Laptop price.

Act:

The agent called the `get_product_price` tool to read the price from the CSV file.

Observe:

The tool returned the Laptop price as **80,000**.

Act:

The agent then called the `multiply_numbers` tool with 80,000 and 2.

Observe:

The tool returned **160,000**.

Final Answer:

The agent used the tool results to give the final answer to the user.

Reflection

LangChain was easier to use because it provided ready-made components for agents, tools, memory, and structured output. Compared with my Day 1 raw-Python agent, I needed less code to manage the agent workflow. Some internal steps are handled automatically by LangChain, so it can sometimes feel like there is some “magic” happening in the background. However, this abstraction makes building an AI agent faster and more organized.

Conclusion

Overall, LangChain provides a useful framework for building AI agents that can use tools, remember conversations, and return structured answers.